Sufficient Conditions for Set Invariance using Multiple Barrier Functions in Geometric Hybrid Dynamical Systems

Piyush P. Jirwankar, Daniel E. Ochoa, and Ricardo G. Sanfelice

***Abstract*— We present sufficient conditions using a vector-valued barrier function, namely a *multiple barrier function*, for strong forward invariance of a closed set for a class of hybrid dynamical systems that evolve on C^1-manifolds. We assume the barrier function to be locally Lipschitz in a neighborhood of the desired closed set, where this notion of Lipschitz continuity is defined without assuming a Riemannian metric on the manifold. This class of barrier functions admits closed sets that may not be otherwise certified to be forward invariant using continuously differentiable and scalar-valued barrier functions. Several examples illustrate the results.**

I. Introduction

Forward invariance of a set implies that all solutions to a dynamical system starting from the set remain in it for all time. This property is fundamental for the study and design of control laws for modern cyber-physical systems, where multiple state constraints need to be satisfied simultaneously to ensure safe operation. Due to the fact that cyber-physical systems have states that evolve *continuously* (for example, position, velocity, etc.) and those that evolve *discretely* (for example, logic variables, timer, etc), these systems are effectively modeled as hybrid dynamical systems [1]. Such modeling couples the continuous and discrete behavior through a combination of constrained differential and difference inclusions.

Barrier functions are commonly used to provide sufficient conditions for forward invariance of a closed set $K \subset \mathbb{R}^n$ (c.f. [2] and the references therein), where the zero-sublevel set of the barrier function is the set K. However, due to multiple nonsmooth constraints that may arise in cyber-physical systems, the structure of the set K may not be captured sufficiently well using a single barrier function. This fact has led to using a *multiple* barrier function [3], which is a vector-valued barrier function such that the intersection of the zero-sublevel sets of its component scalar functions results in the set K. The convenience and generality of such multiple barrier functions has resulted in its increased use in recent literature [4], [5]. Sufficient conditions using multiple barrier functions for forward invariance of a closed set for hybrid dynamical systems have been presented in [3], where the state of the underlying hybrid dynamical system is assumed to evolve on $\mathbb{R}^n$.

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However, a large class of cyber-physical systems have states that evolve on general C^1-manifolds and not in $\mathbb{R}^n$. Using recent advances in topological and coordinate-independent modeling of hybrid systems [6], [7], such systems can be modeled as *geometric hybrid dynamical systems* on C^1-manifolds. Whether the sufficient conditions in [3] for forward invariance of a set using multiple barrier functions hold for this class of systems remains unknown. A major challenge in addressing this question is obtaining pointwise decrease conditions that uniformly bound the change of the barrier function candidate along solutions on the manifold. Doing so requires the ability to compute the Clarke generalized directional derivative of a locally Lipschitz barrier function on a C^1-manifold, with the directions belonging to the contingent cone of some set. The Euclidean tools prove insufficient to define these notions on manifolds.

The nonsmooth and set-valued analysis tools on C^1-manifolds from [6] address the above challenges by providing geometric, coordinate-independent definitions of the above notions, thereby allowing the extension of the results in [3] from the Euclidean setting to geometric hybrid dynamical systems. To this end, the contributions of this paper are as follows:

1) Using the definition of locally Lipschitz functions between manifolds from [6], we extend [3], [8, Ch. 11] and define a nonsmooth multiple barrier function candidate such that the intersection of the zero-sublevel sets of each of its component functions results in the set K.
2) The above candidate function is then used to certify forward (pre-)invariance of a closed set using decrease conditions similar to those presented in [8, Thm. 11.14] and [3, Thm. 1] for the Euclidean case.
3) Several examples demonstrate the applicability of the results.

The remainder of this paper is organized as follows. Section II introduces required preliminaries. Section III recalls the geometric hybrid dynamical systems framework from [6], [7]. Section IV presents nonsmooth analysis notions and results, and Section V defines the desired set invariance notions. Finally, sufficient conditions for forward invariance are presented in Section VI. Due to space constraints, proofs will be presented elsewhere.

II. Preliminaries

A. Notation

We use $\mathbb{R}$ and $\mathbb{R}_{\geq 0}$ to denote the set of real and nonnegative real numbers, respectively. The set of natural numbers, including 0, is denoted by $\mathbb{N}$. A function $f : \mathbb{R}^m \to \mathbb{R}^n$ is

said to be a C^k-function if it has k continuous derivatives. A set-valued map $F : X \rightrightarrows Y$ associates every element in X with a subset of Y. The domain, range, and graph of F are defined as $\operatorname{dom} F := \{x : F(x) \neq \varnothing\}$, $\operatorname{rge} F := \{y : \exists x \text{ s.t. } y \in F(x)\}$, and $\operatorname{gph} F := \{(x, y) : y \in F(x)\}$, respectively. Given a nonempty set $S \subset \mathbb{R}^n$, the tangent cone $\mathrm{T}_S(x)$ to S at $x \in S$ is the set of all vectors $w \in \mathbb{R}^n$ for which there exist sequences $x_i \in S$, $\tau_i > 0$ with $x_i \to x$, $\tau_i \to 0$, and $w = \lim_{i\to\infty} (x_i - x)/\tau_i$.

Given a topological space $(\mathcal{X}, \mathcal{T})$, for a set $S \subset \mathcal{X}$, we write $\overline{S}$ for its closure and $\operatorname{int} S$ for its interior. A set $S \subset \mathcal{X}$ is *compact* if each open cover of S has a finite subcover, and it is *precompact* if $\overline{S}$ is compact. Given $r \in \mathbb{R}$ and a function $f : \mathcal{X} \to \mathbb{R}$, the r-sublevel set of f is denoted by $L_f(r) := \{x \in \mathcal{X} : f(x) \leq r\}$.

B. Differentiable Manifolds

An n-dimensional topological manifold $\mathcal{M}$ is a second-countable Hausdorff space that is locally Euclidean of dimension n. A coordinate chart of $\mathcal{M}$ at x is a pair (U, φ) where $x \in U \subset \mathcal{M}$, with U an open set and $\varphi : U \to \varphi(U) \subset \mathbb{R}^n$ a homeomorphism. Two charts (U, φ) and (V, ψ) of $\mathcal{M}$ are C^k-compatible if $U \cap V = \varnothing$ or $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is a C^k-diffeomorphism. A C^k-atlas of $\mathcal{M}$ is one whose charts are C^k-compatible. A C^k-manifold $\mathcal{M}$ is a topological manifold endowed with a maximal C^k-atlas. The tangent space to $\mathcal{M}$ at $x \in \mathcal{M}$ is denoted by $T_x\mathcal{M}$, and the tangent bundle of $\mathcal{M}$ is denoted by $T\mathcal{M} := \{(x, v) : x \in \mathcal{M}, v \in \mathrm{T}_x\mathcal{M}\}$. A Riemannian manifold is a pair $(\mathcal{M}, g_{\mathcal{M}})$, where $\mathcal{M}$ is a C^k-manifold and $g_{\mathcal{M}}$ is a Riemannian metric. See [9] for more details on differential and Riemannian manifolds.

We consider two examples of manifolds. First, consider the unit circle $\mathbb{S}^1$ as the quotient space $[0, 1]/\{0, 1\}$ with the equivalence relation $0 \sim 1$. We equip $\mathbb{S}^1$ with a metric, causing it to be a smooth Riemannian manifold. Moreover, $\mathrm{T}\mathbb{S}^1$ is a smooth manifold [10, Prop. 3.18]. Elements of $\mathrm{T}\mathbb{S}^1$ are written as $([z_1], [z_2])$, where $[z_1] \in \mathbb{S}^1$, $[z_2] \in \mathrm{T}_{[z_1]}\mathbb{S}^1$, and $[\cdot]$ denotes the corresponding equivalence class.

Second, consider the set of two-dimensional rotation matrices $\mathrm{SO}(2) := \{R \in \mathbb{R}^{2\times 2} : RR^\top = R^\top R = I,\ \det R = 1\}$, which is a Lie group with identity $I \in \mathrm{SO}(2)$. Then, $\mathrm{SO}(2)$ is also a Riemannian manifold with the standard bi-invariant metric $g_{\mathrm{SO}(2)}$ induced by the group structure. The distance between $R_1, R_2 \in \mathrm{SO}(2)$ is denoted by $|R_1|_{R_2} := d_{g_{\mathrm{SO}(2)}}(R_1, R_2)/\pi$, where $d_{g_{\mathrm{SO}(2)}} : \mathrm{SO}(2)\times\mathrm{SO}(2) \to \mathbb{R}_{\geq 0}$ is the geodesic distance associated with the bi-invariant metric. Note that $d_{g_{\mathrm{SO}(2)}}(R_1, R_2) \in [0, 1]$ for each $R_1, R_2 \in \mathrm{SO}(2)$.

III. Geometric Hybrid Dynamical Systems

Let $\mathcal{M}$ be a C^1-manifold. We recall geometric hybrid dynamical systems from [6] described by the hybrid inclusion

$$\mathcal{H} : \begin{cases} \dot{x} \in F(x) & x \in C \\ x^+ \in G(x) & x \in D \end{cases}, \tag{1}$$

where $x \in \mathcal{M}$ is the state of the system, $C \subset \mathcal{M}$ is the flow set, $D \subset \mathcal{M}$ is the jump set, and $G : \mathcal{M} \rightrightarrows \mathcal{M}$ is the jump map. The flow map is a set-valued map satisfying $F(x) \subset T_x\mathcal{M}$ for each $x \in C$, and is denoted as $F : \mathcal{M} \rightrightarrows T\mathcal{M}$. We use $\mathcal{H} = (C, F, D, G, \mathcal{M})$ to represent a geometric hybrid system using this data.

Solutions to $\mathcal{H}$ are defined on *hybrid time domains*. A set $E \subset \mathbb{R}_{\geq 0} \times \mathbb{N}$ is a hybrid time domain if it is the union of compact hybrid time domains E_j with the property that $E_0 \subset E_1 \subset E_2 \subset \ldots \subset E_j \ldots$. A set $E \subset \mathbb{R}_{\geq 0} \times \mathbb{N}$ is a compact hybrid time domain if there exists $J \in \mathbb{N}$ such that $E = \bigcup_{j=0}^{J} [t_j, t_{j+1}] \times \{j\}$, for some finite sequence of times $\{t_j\}_{j=0}^{J+1}$ satisfying $0 = t_0 \leq t_1 \leq t_2 \leq \ldots \leq t_J \leq t_{J+1}$.

A function $\phi : \operatorname{dom}\phi \to \mathcal{M}$ is a hybrid arc if $\operatorname{dom}\phi$ is a hybrid time domain and if, for each $j \in \mathbb{N}$, the function $t \mapsto \phi(t, j)$ is locally absolutely continuous[1] on the interval $I^j := \{t : (t, j) \in \operatorname{dom}\phi\}$.

Definition 3.1 (Solution to a hybrid system)**.** *A hybrid arc ϕ is a solution to the geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ if $\phi(0, 0) \in \overline{C} \cup D$ and the following conditions hold:*

1) For each $j \in \mathbb{N}$ such that I^j has nonempty interior,

$$\begin{aligned} \phi(t, j) &\in C && \text{for all} && t \in \operatorname{int} I^j \\ \frac{d\phi}{dt}(t, j) &\in F(\phi(t, j)) && \text{for almost all} && t \in I^j \end{aligned};$$

2) For each $(t, j) \in \operatorname{dom}\phi$ such that $(t, j + 1) \in \operatorname{dom}\phi$,

$$\phi(t, j) \in D \qquad \text{and} \qquad \phi(t, j + 1) \in G(\phi(t, j)).$$

For a solution ϕ to $\mathcal{H}$, define $\sup_t \operatorname{dom}\phi := \{t : \exists j \text{ s.t. } (t, j) \in \operatorname{dom}\phi\}$, $\sup_j \operatorname{dom}\phi := \{j : \exists t \text{ s.t. } (t, j) \in \operatorname{dom}\phi\}$, and $\operatorname{length}(\operatorname{dom}\phi) := \sup_t \operatorname{dom}\phi + \sup_j \operatorname{dom}\phi$.

A solution ϕ is *nontrivial* if $\operatorname{dom}\phi$ contains at least two points. It is *complete* if $\operatorname{length}(\operatorname{dom}\phi) = \infty$. It has a *finite escape time* if, for each compact set $S \subset \mathcal{M}$, there exists $T > 0$ such that $\phi(t, j) \notin S$ for each $(t, j) \in \operatorname{dom}\phi$ with $t + j \geq T$. It is *maximal* if no other solution ψ to $\mathcal{H}$ exists such that $\operatorname{dom}\phi$ is a proper subset of $\operatorname{dom}\psi$ and $\phi(t, j) = \psi(t, j)$ for all $(t, j) \in \operatorname{dom}\phi$. The set of maximal solutions to $\mathcal{H}$ from a nonempty set $K \subset \mathcal{M}$ is denoted by $\mathcal{S}_{\mathcal{H}}(K)$.

Now, we illustrate (1) in two examples of systems that model kinematics and dynamics on manifolds.

Example 3.2 (Constrained kinematics on $\mathrm{SO}(2)$)**.** *Consider the geometric hybrid dynamical system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ with $x := (R, q) \in \mathcal{M}$, where $R \in \mathrm{SO}(2)$, $q \in \{-1, 1\}$, and $\mathcal{M} := \mathrm{SO}(2) \times \{-1, 1\}$, as*

$$C := \bigcup_{q\in\{-1,1\}} C_q \times \{q\},$$
$$D := \bigcup_{q\in\{-1,1\}} \left(\overline{\mathrm{SO}(2) \setminus C_q}\right) \times \{q\},$$

where, for constants $0 < c_{-1} < c_1 < 1$, $C_1 := \{R : |R|_I \leq c_1\}$ and $C_{-1} := \{R : |R|_I \geq c_{-1}\}$. Note that $C \cup D = \mathcal{M}$. For each $x \in \mathcal{M}$, the maps F and G are given by

$$F(x) := (q\nabla V(R), 0) \qquad G(x) := (R, -q),$$

[1] A function $\gamma : \mathbb{R} \supset I \to \mathcal{M}$ is said to be locally absolutely continuous if, for each C^1-smooth function $\psi : \mathcal{M} \to \mathbb{R}$, the composition $\psi \circ \gamma : I \to \mathbb{R}$ is locally absolutely continuous [11, Section A.2.1].

where $V(R) \coloneqq |R|_I^2$ for each $R \in \mathrm{SO}(2)$. This geometric hybrid system $\mathcal{H}$ models the kinematics of a planar pick-and-place robotic arm under the action of a hybrid control law.

Example 3.3 (Pendulum with impacts)**.** *Consider a pendulum rotating on the unit circle $\mathbb{S}^1$, and recall that $\mathbb{S}^1$ is modeled as a quotient space. The unit circle contains a connected wall $\mathcal{W}$ of thickness 2α, $\alpha \in (0, 1/2)$, defined as $\mathcal{W} \coloneqq ([0, \alpha] \cup [1 - \alpha, 1])/\{0, 1\}$. The position $[z_1] \in \mathbb{S}^1$, with $z_1 \in [0, 1]$, and the velocity $[z_2] \in \mathrm{T}_{[z_1]}\mathbb{S}^1$, with $z_2 \in \mathbb{R}$, constitute the state $x \coloneqq ([z_1], [z_2]) \in \mathrm{T}\mathbb{S}^1$ of the pendulum.*

The dynamics of the pendulum evolve continuously under gravitational potential $U : \mathbb{S}^1 \to \mathbb{R}_{\geq 0}$, defined as $U([z_1]) \coloneqq \frac{a}{2\pi}(1 - \cos(2\pi z_1))$ for each $z_1 \in [0, 1]$, where $a > 0$, until it hits the wall $\mathcal{W}$, after which the velocity of the pendulum changes direction and is scaled by an unknown factor $\lambda \in [0, 1]$. These dynamics are described by the hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ with $\mathcal{M} \coloneqq \mathrm{T}\mathbb{S}^1$, where

$$
\begin{aligned}
C &\coloneqq \{x \in \mathcal{M} : [z_1] \in \overline{\mathbb{S}^1 \setminus \mathcal{W}}, [z_2] \in \mathrm{T}_{[z_1]}\mathbb{S}^1\},\\
F(x) &\coloneqq \mathfrak{S}(x) + \mathrm{vlft}_x(-\nabla U([z_1]) - b[z_2]) \qquad \forall x \in C,\\
D &\coloneqq \{x \in \mathcal{M} : [z_1] \in \partial\mathcal{W}, [z_2] \in \mathrm{T}^{\mathbb{S}^1}_{\mathcal{W}}([z_1])\},\\
G(x) &\coloneqq \{([z_1], [-\lambda z_2]) : \lambda \in [0, 1]\} \qquad \forall x \in D,
\end{aligned}
$$

and $b > 0$. Note that $G(D) \subset C$. The map $\mathfrak{S} : \mathcal{M} \to \mathrm{T}\mathcal{M}$ denotes the "geodesic spray" of $\mathcal{M}$, which is a vector field on $\mathcal{M} = \mathrm{T}\mathbb{S}^1$ whose integral curves, when projected using the canonical tangent bundle projection map $\Pi : \mathrm{T}\mathbb{S}^1 \to \mathbb{S}^1$, are geodesics on $\mathbb{S}^1$. The map $\mathrm{vlft}_x : \mathrm{T}_{\Pi(x)}\mathbb{S}^1 \to \mathrm{T}_x\mathrm{T}\mathbb{S}^1$ denotes the vertical lift at $x \in \mathcal{M}$ that maps external forces to the tangent space of the tangent bundle. Figure 2 illustrates this fact. Together, the vertical lift and the geodesic spray lift second-order dynamics on $\mathbb{S}^1$ to first-order dynamics on $\mathcal{M}$. We refer the reader to [11, Remark 3.103, p. 113] and [11, Definition 3.73, p. 96] for details.

We present the following regularity conditions on the data of the geometric hybrid system $\mathcal{H}$. These conditions impart desirable sequential properties to the solution set of $\mathcal{H}$; see [7], and provide checkable conditions for the existence of nontrivial solutions to $\mathcal{H}$; see [6]. These conditions will be used in the forthcoming Corollary 6.7.

Definition 3.4 (Geometric Hybrid Basic Conditions [6, Def. 3])**.** *The system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ is said to satisfy* geometric hybrid basic conditions *if the following hold:*

1) *$\mathcal{M}$ is a C^1-manifold;*
2) *C and D are closed subsets of $\mathcal{M}$;*
3) *$F : \mathcal{M} \rightrightarrows \mathrm{T}\mathcal{M}$ is outer semicontinuous and locally precompact[2] relative to C, $C \subset \mathrm{dom}\, F$, and $F(x)$ is convex for all $x \in C$;*
4) *$G : \mathcal{M} \rightrightarrows \mathcal{M}$ is outer semicontinuous and locally precompact relative to D, and $D \subset \mathrm{dom}\, G$.*

[2]The map $F : \mathcal{M} \rightrightarrows \mathrm{T}\mathcal{M}$ is outer semicontinuous relative to C if the map $\widetilde{F} : \mathcal{M} \rightrightarrows \mathrm{T}\mathcal{M}$, defined as $\widetilde{F}(x) \coloneqq F(x)$ for each $x \in C$ and $\widetilde{F}(x) \coloneqq \varnothing$ otherwise, has closed graph. The map F is locally precompact relative to C if at each $x \in C$, there exists a neighborhood $\mathcal{U}$ of x such that $\widetilde{F}(\mathcal{U})$ is precompact.

IV. Nonsmooth Scalar Functions and Their Generalized Derivatives

In this section, we define the following notion of locally Lipschitz functions between C^1-manifolds.

Definition 4.1 (Locally Lipschitz function)**.** *Given two C^1-manifolds $\mathcal{M}$ and $\mathcal{N}$, a function $f : \mathcal{M} \to \mathcal{N}$ is locally Lipschitz if, for each $x \in \mathcal{M}$, there exist a coordinate chart (U, φ) on $\mathcal{M}$ at x and a coordinate chart (W, ψ) on $\mathcal{N}$ at $f(x)$ such that the function $\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(W)) \to \psi(W)$ is locally Lipschitz in the Euclidean sense.*

The above definition is different from the one presented in [6], where $f : \mathcal{M} \to \mathcal{N}$ is locally Lipschitz if, for each pair of coordinate charts (U, φ) on $\mathcal{M}$ and (V, ψ) on $\mathcal{N}$, the map $\psi \circ f \circ \varphi^{-1}$ is locally Lipschitz in the Euclidean sense. The reformulated definition above mimics the structure of the notion of locally Lipschitz functions between Euclidean spaces by requiring existence of (rather than requiring the property to hold for all) open neighborhoods at each point $x \in \mathcal{M}$ and $f(x) \in \mathcal{N}$, and specializes an existing notion [12, Definition 3] of Hölder continuous functions between manifolds.

Given a locally Lipschitz function $B : \mathrm{dom}\, B \to \mathbb{R}$, we can compute its (Clarke) generalized directional derivative at a point $x \in \mathcal{M}$ in the direction of $v \in \mathrm{T}_x\mathcal{M}$ using local coordinate charts. We recall this definition from [6] below.

Definition 4.2 ((Clarke) Generalized directional derivative)**.** *The (Clarke) generalized directional derivative of a locally Lipschitz function $B : \mathrm{dom}\, B \to \mathbb{R}$, $\mathrm{dom}\, B \subset \mathcal{M}$, at $x \in \mathcal{M}$ in the direction $v \in \mathrm{T}_x\mathcal{M}$ is denoted by $B^\circ(x, v)$ and defined as*

$$
B^\circ(x, v) \coloneqq \limsup_{\substack{y \to x \\ t \searrow 0}} \frac{\widetilde{B}(\varphi(y) + t\mathrm{d}\varphi_x(v)) - \widetilde{B}(\varphi(y))}{t} \tag{2}
$$

where (U, φ) is a chart at x and $\widetilde{B} \coloneqq B \circ \varphi^{-1}$; recall that $\mathrm{d}\varphi_x : \mathrm{T}_x\mathcal{M} \to \mathbb{R}^{\dim \mathcal{M}}$ denotes the differential of φ at x.

This notion of directional derivatives enables the computation of the maximum rate of change of B along solutions to $\mathcal{H}$ as described in forthcoming Definition 4.4. To do so, we recall from [6, Def. 4] the notion of a tangent cone to $S \subset \mathcal{M}$ at $x \in \mathcal{M}$.

Definition 4.3 (Tangent cone to subsets of manifolds)**.** *Given a nonempty subset $S \subset \mathcal{M}$, the tangent cone $\mathrm{T}^{\mathcal{M}}_S(x)$ at $x \in S$ is defined as $\mathrm{T}^{\mathcal{M}}_S(x) \coloneqq (\mathrm{d}\varphi_x)^{-1}(\mathrm{T}_{\varphi(S\cap U)}(\varphi(x)))$, where (U, φ) is any coordinate chart of $\mathcal{M}$ at x and $\mathrm{T}_{\varphi(S\cap U)}(\varphi(x))$ denotes the Euclidean tangent cone to the set $\varphi(S \cap U) \subset \mathbb{R}^n$ at the point $\varphi(x)$.*

Definition 4.4 ($\dot{B}$ and B^+)**.** *Given a geometric hybrid dynamical system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ and a locally Lipschitz function $B : \mathrm{dom}\, B \to \mathbb{R}$ with $\overline{C} \cup D \cup G(D) \subset \mathrm{dom}\, B$,*

- *the largest change of B along flows is given by*

$$
\dot{B}(x) \coloneqq \sup_{v \in F(x) \cap \mathrm{T}^{\mathcal{M}}_C(x)} B^\circ(x, v) \quad \forall x \in C,
$$

- *the largest value of B after jumps is given by*

$$B^+(x) \coloneqq \sup_{g \in G(x)} B(g) \qquad \forall x \in D.$$

The above characterization of the largest change of B during flows and at jumps along solutions to $\mathcal{H}$ is used in forthcoming Section VI to obtain invariance certificates.

V. Forward Pre-Invariance Notions for Geometric Hybrid Systems

Consider a geometric hybrid dynamical system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ and a nonempty set $K \subset C \cup D$. Following [3], [8, Ch. 11], we define the following notions of (strong) forward invariance of the set K for the geometric hybrid dynamical system $\mathcal{H}$.

Definition 5.1 (Forward pre-invariance)**.** *A set $K \subset \mathcal{M}$ is said to be* forward pre-invariant *for $\mathcal{H}$ if each solution ϕ to $\mathcal{H}$ from K satisfies $\operatorname{rge}\phi \subset K$. The set K is said to be* forward invariant *if, in addition, each maximal solution $\phi \in \mathcal{S}_{\mathcal{H}}(K)$ is also complete.*

Example 5.2 (Reachable set is forward pre-invariant)**.** *For each $\xi \in \overline{C} \cup D$, the reachable set $\mathcal{R}_{\mathcal{H}}(\xi)$, defined as*

$$\mathcal{R}_{\mathcal{H}}(\xi) \coloneqq \{\phi(t,j) : \phi \in \mathcal{S}_{\mathcal{H}}(\xi), (t,j) \in \operatorname{dom}\phi\}, \qquad (3)$$

is forward pre-invariant for $\mathcal{H}$. Indeed, proceeding by contradiction, suppose there exist $\zeta \in \mathcal{R}_{\mathcal{H}}(\xi)$, $\psi \in \mathcal{S}_{\mathcal{H}}(\zeta)$, and $(s,k) \in \operatorname{dom}\psi$ such that $\psi(s,k) \notin \mathcal{R}_{\mathcal{H}}(\xi)$. From (3) and the fact that $\zeta \in \mathcal{R}_{\mathcal{H}}(\xi)$, there exists $\gamma \in \mathcal{S}_{\mathcal{H}}(\xi)$ such that $\gamma(s',k') = \zeta$ for some $(s',k') \in \operatorname{dom}\gamma$. Then, constructing a solution ϕ to $\mathcal{H}$ from ξ such that $\phi(t,j) \coloneqq \gamma(t,j)$ for each $(t,j) \in \operatorname{dom}\gamma$ with $t + j \leq s' + k'$ and $\phi(t,j) \coloneqq \psi(t-s', j-k')$ for each $(t-s', j-k') \in \operatorname{dom}\psi$, we observe that $\phi(s+s', k+k') \notin \mathcal{R}_{\mathcal{H}}(\xi)$, which is a contradiction.

Example 5.3 (Basin of attraction is forward invariant)**.** *Suppose that a compact set $\mathcal{A}$ is asymptotically stable[3] for $\mathcal{H} = (C, F, D, G, \mathcal{M})$. Let $\mathcal{B}(\mathcal{A}) \subset \mathcal{M}$ denote the basin of attraction of $\mathcal{A}$ for $\mathcal{H}$; in particular, $\mathcal{B}(\mathcal{A})$ is the set of all points in $\mathcal{M}$ from where each maximal solution to $\mathcal{H}$ is complete and converges to $\mathcal{A}$. Then, using arguments similar to those in Example 5.2, $\mathcal{B}(\mathcal{A})$ is forward invariant for $\mathcal{H}$.*

VI. Sufficient Conditions for Strong Forward Pre-Invariance of a Set

Analogous to the role of Lyapunov functions in certifying asymptotic stability of a set, barrier functions are employed to establish forward invariance of a set. Given a nonempty set $K \subset C \cup D$, following [3], the key idea is to use a continuous, vector-valued function $B : \operatorname{dom} B \to \mathbb{R}^N$ (that is locally Lipschitz on an open neighborhood outside of K) to certify forward pre-invariance of K for $\mathcal{H}$. The function B has N components given by functions $B_i : \operatorname{dom} B \to \mathbb{R}$, where $i \in \mathcal{I}$, with $\mathcal{I}$ denoting a finite index set with $\operatorname{card}(\mathcal{I}) = N$. For each $i \in \mathcal{I}$, denoting the zero-sublevel set of the component function $B_i : \operatorname{dom} B_i \to \mathbb{R}$ by

$$K_i \coloneqq \{x \in C \cup D : B_i(x) \leq 0\}, \qquad (4)$$

the set K is equal to the intersection of K_i's over $i \in \mathcal{I}$.

We formalize the above construction in the following definition, which is an extension of [8, Def. 11.8] to the geometric setting.

Definition 6.1 (Multiple barrier function candidate)**.** *A continuous function $B : \operatorname{dom} B \to \mathbb{R}^N$ is said to define a multiple barrier function candidate with respect to a nonempty set $K \subset \mathcal{M}$ for the geometric hybrid dynamical system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ if the following conditions hold:*

(i) $\overline{C} \cup D \cup G(D) \subset \operatorname{dom} B$ and $K \subset C \cup D$;

(ii) for each $i \in \mathcal{I}$, there exists an open set U_i with $M_i \subset U_i$ such that B_i is locally Lipschitz on $(U_i \setminus K_i) \cap \overline{C}$, where $M_i \coloneqq \{x \in \partial K : B_i(x) = 0\}$;

(iii) $K = \bigcap_{i \in \mathcal{I}} K_i$, where the set K_i is defined in (4).

Additionally, if $N = 1$, we call B a scalar barrier function candidate with respect to K for $\mathcal{H}$.

Remark 6.2. *We highlight that the open sets U_i, $i \in \mathcal{I}$, in Definition 6.1 are not related, unless otherwise specified, to coordinate charts on $\mathcal{M}$.*

We note that the regularity of the barrier function candidate is required in a neighborhood outside K because the forward invariance property only depends on the vector field immediately outside K; c.f. [13, Chapter 4.6]. This fact is observed in the forthcoming Theorem 6.5.

When K is described by multiple smooth constraints, the construction in Definition 6.1 enables the use of a smooth multiple barrier function candidate whose component functions B_i, $i \in \mathcal{I}$, represent these constraints. This is in contrast to employing a scalar barrier function candidate $B' : \operatorname{dom} B' \to \mathbb{R}$, which could instead be defined as

$$B'(x) \coloneqq \max_{i \in \mathcal{I}} B_i(x) \qquad \forall x \in \operatorname{dom} B,$$

but generally fails to preserve the smoothness properties of the functions B_i. A similar observation has been noted in [3].

Example 6.3 (Constrained kinematics on $\mathrm{SO}(2)$, revisited)**.** *Recall Example 3.2. For each $i \in \mathcal{I} \coloneqq \{-1, 1\}$, define $B_i : \mathcal{M} \to \mathbb{R}$ as*

$$B_i(x) \coloneqq i\left(c_{-i}^2 - V(R)\right) \qquad \forall x \in \mathcal{M}. \qquad (5)$$

Note from (4) that $K_{-1} = \{x : |R|_I \leq c_1\}$ and $K_1 = \{x : |R|_I \geq c_{-1}\}$. Define $K \coloneqq K_{-1} \cap K_1$ and note that $K \neq \varnothing$ as $c_{-1} < c_1$. Note also that $K = \widetilde{K} \times \{-1, 1\}$ with $\widetilde{K} \coloneqq \{R \in \mathrm{SO}(2) : c_{-1} \leq |R|_I \leq c_1\}$. Defining $B : \mathcal{M} \to \mathbb{R}^2$ as

$$B(x) \coloneqq [B_{-1}(x) \;\; B_1(x)]^\top \qquad \forall x \in \mathcal{M},$$

and noting that B is C^1 on $\mathcal{M}$, the function B is a multiple barrier function candidate with respect to K for $\mathcal{H}$. While a function $\widetilde{B} : \mathcal{M} \to \mathbb{R}$ defined as $\widetilde{B}(x) \coloneqq B_q(R)$ for each $x \in \mathcal{M}$ acts as a C^1 scalar barrier function candidate with

[3] In particular, $\mathcal{A}$ is stable and pre-attractive for $\mathcal{H}$, and each maximal solution to $\mathcal{H}$ is complete; see [6] for more details.

respect to K for $\mathcal{H}$, we define the multiple barrier function candidate above for illustrative purposes.

Example 6.4 (Pendulum with impacts, revisited)**.** *Recall Example 3.3. Consider the energy function $V : C \cup D \to \mathbb{R}_{\geq 0}$ such that, for each $x \in C \cup D$, $V(x) \coloneqq U([z_1]) - U(\mathcal{A}) + \frac{1}{2} z_2^2$, where $\mathcal{A} \coloneqq \partial\mathcal{W} \times \{0\} \subset \mathcal{M}$ so that V is positive definite with respect to $\mathcal{A}$. We note that V is locally Lipschitz on $C \cup D$. Fix $r > 0$ and define $K \coloneqq L_V(r)$ as the r-sublevel set of V. Then, the function $B : \mathcal{M} \to \mathbb{R}$, defined as*

$$B(x) \coloneqq V(x) - r \qquad \forall x \in \mathcal{M}, \tag{6}$$

is a scalar barrier function candidate with respect to K for $\mathcal{H}$.

A. Sufficient Conditions for Forward (pre-)invariance

We present the main result which generalizes [3, Theorem 1] to the geometric setting.

Theorem 6.5 (Forward pre-invariance using a multiple barrier function)**.** *Given a geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ as in (1) and a closed set $K \subset \mathcal{M}$, suppose that $B : \operatorname{dom} B \to \mathbb{R}^N$ is a multiple barrier function candidate with respect to K for $\mathcal{H}$. For each $i \in \mathcal{I}$, let U_i be an open neighborhood of M_i as in Definition 6.1. Then, the set K is forward pre-invariant for $\mathcal{H}$ if, for each $i \in \mathcal{I}$,*

$$\dot{B}_i(x) \leq 0 \qquad \forall x \in (U_i \backslash K_i) \cap C \tag{7a}$$

$$B_i^+(x) \leq 0 \qquad \forall x \in D \cap K_i \tag{7b}$$

$$G(D \cap K) \subset C \cup D \tag{7c}$$

We note that nonstrict decrease during flows in (7a) is required in a neighborhood outside K because the forward invariance property only depends on the vector field immediately outside K; c.f. [13, Ch. 4.6]. This fact motivates sufficient regularity of B_i, $i \in \mathcal{I}$, only on $(U_i \setminus K_i) \cap \overline{C}$.

Next, we relate barrier certificates to Lyapunov certificates with nonstrict decrease; see [7] for a definition of a Lyapunov function candidate for $\mathcal{H}$.

Corollary 6.6. *Consider a geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ such that $C \cup D = \mathcal{M}$. Given a nonempty, closed set $\mathcal{A} \subset \mathcal{M}$, suppose there exists a locally Lipschitz function $V : \mathcal{M} \to \mathbb{R}_{\geq 0}$ with $V \in \mathcal{PD}(\mathcal{A})$ that satisfies*

$$\dot{V}(x) \leq 0 \qquad x \in C, \tag{8}$$

$$V^+(x) - V(x) \leq 0 \qquad x \in D. \tag{9}$$

Then, for each $r \geq 0$, the r-sublevel set $L_V(r) \coloneqq \{x \in \mathcal{M} : V(x) \leq r\}$ is forward pre-invariant for $\mathcal{H}$.

Finally, requiring completeness of maximal solutions from K, in addition to the conditions of Theorem 6.5, strengthens the certificate therein to forward invariance of K for $\mathcal{H}$.

Corollary 6.7. *Consider a geometric hybrid dynamical system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ that satisfies the geometric hybrid basic conditions in Definition 3.4. Then, a closed set $K \subset \mathcal{M}$ is forward invariant for $\mathcal{H}$ if* (7a)-(7c) *hold, for each $\xi \in (K \cap \partial C) \setminus D$, there exists an open neighborhood U of ξ such that $F(x) \cap \mathrm{T}_C^{\mathcal{M}}(x) \neq \varnothing$ holds for each $x \in U \cap C$, and at least one of the following conditions holds:*

1) *$K \cap C$ is compact;*
2) *The constrained differential inclusion $\dot{x} \in F(x) \quad x \in C$ does not have a maximal solution with a finite escape time.*

Remark 6.8. *In contrast to [3, Thm. 1], Theorem 6.5 does not require $\mathcal{H}$ to satisfy the geometric hybrid basic conditions. This condition is only invoked in Corollary 6.7 to exploit [6, Prop. 2], where sufficient conditions for existence and completeness of solutions under Definition 3.4 are presented.*

Remark 6.9. *Given a hybrid plant with control input u, the result in Theorem 6.5 can be used to design safe control laws such that the set K is rendered forward pre-invariant in the presence of obstacles for the resulting closed-loop system. The design of such a controller for constrained differential inclusions in the Euclidean space; i.e., when $D = \varnothing$ and $\mathcal{M} = \mathbb{R}^n$, is presented in [14].*

Example 6.10 (Constrained kinematics on $\mathrm{SO}(2)$, revisited)**.** *We show that the system $\mathcal{H}$ in Examples 3.2 and 6.3 is such that, in the presence of obstacles, the orientation R of the robotic arm stays in the* safe set *$\widetilde{K}$. For each $i \in \mathcal{I}$, let $U_i \coloneqq \mathcal{M}$ so that U_i is an open neighborhood of $\{x \in \partial K : B_i(x) = 0\}$. Then, for each $i \in \mathcal{I}$,*

$$(U_i \setminus K_i) \cap C = \{R : i(|R|_I - c_{-i}) < 0\} \times \{i\}. \tag{10}$$

Consequently, for each $i \in \mathcal{I}$ and each $x \in (U_i \setminus K_i) \cap C$,

$$\dot{B}_i(x) = g_{\mathcal{M}}(\nabla B_i(x), F(x))$$

where $g_{\mathcal{M}}$ is the Riemannian metric induced on $\mathcal{M}$ using the Riemannian metric $g_{\mathrm{SO}(2)}$ on $\mathrm{SO}(2)$. Then, for each $i \in \mathcal{I}$ and each $x \in (U_i \setminus K_i) \cap C$,

$$\begin{aligned}
\dot{B}_i(x) &= \mathrm{d}(B_i)_x(F(x)) \\
&= g_{\mathcal{M}}(\nabla B_i(x), F(x)) \\
&= g_{\mathrm{SO}(2)}(-i\nabla V(R), q\nabla V(R)) \\
&= -iq\ g_{\mathrm{SO}(2)}(\nabla V(R), \nabla V(R)) \\
&= -i^2\ g_{\mathrm{SO}(2)}(\nabla V(R), \nabla V(R)) \\
&\leq 0
\end{aligned}$$

where the second equality holds by the definition of Riemannian metric, and the last inequality is due to the fact that, using (10)*, $q = i$ for each $i \in \mathcal{I}$ and each $x \in (U_i \setminus K_i) \cap \overline{C}$, and that $g_{\mathrm{SO}(2)}(v, v) \geq 0$ for each $(R, v) \in \mathrm{TSO}(2)$. Therefore,* (7a) *holds.*

Next, note that $B_i(G(x)) = B_i(x) \leq 0$ for each $x \in D \cap K_i$ and each $i \in \mathcal{I}$, where the inequality follows from (4)*. Therefore,* (7b) *holds. Finally,* (7c) *holds trivially as $C \cup D = \mathcal{M}$. As a result, Theorem 6.5 certifies that K is, indeed, forward pre-invariant for $\mathcal{H}$. This result is illustrated in Figure 1.*

To prove forward invariance of K for $\mathcal{H}$, we note that $\mathcal{H}$ satisfies geometric hybrid basic conditions in Definition 3.4. Note that $(K \cap \partial C) \setminus D$ is an open set, so, for each $\xi \in (K \cap \partial C) \setminus D$, there exists an open neighborhood such that, for each point x in the neighborhood, $F(x) \subset \mathrm{T}_C^{\mathcal{M}}(x) = \mathrm{T}_x\mathcal{M}$. Then, as $K \cap C$ is compact due to compactness of $\mathcal{M}$, Corollary 6.7 certifies that K is forward invariant for $\mathcal{H}$.

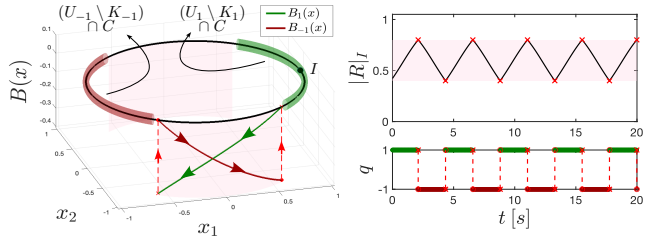


Fig. 1: *Left:* A representation of $\mathrm{SO}(2)$ as the unit circle in $\mathbb{R}^2$ (in black). The pink region on the circle denotes the set $\widetilde{K} \subset \mathrm{SO}(2)$ where $B_i(x) \leq 0$, $i \in \mathcal{I}$. The shaded brown and green regions denote the projection of $(U_i \setminus K_i) \cap C$ onto $\mathrm{SO}(2)$. For $\phi \in \mathcal{S}_{\mathcal{H}}(K)$, the brown and green arrows denote $(B_i \circ \phi)$, $i \in \mathcal{I}$; both remaining in $\mathbb{R}_{\leq 0}$, so the solution ϕ stays in K. *Right:* Evolution of $|R(t,j)|_I$ and $q(t,j)$, where $\phi = (R, q)$, projected onto the t-axis. The dotted red lines indicate a jump in the solution. With $c_{-1} = 0.3$ and $c_1 = 0.7$, the pink region indicates $\operatorname{rge} R \in \widetilde{K}$.

Therefore, as $K = \widetilde{K} \times \{-1, 1\}$, the orientation of the robotic arm stays in $\widetilde{K}$.

Example 6.11 (Pendulum with impacts, revisited)**.** *Recall Examples 3.3 and 6.4. Using* (6) *and the analysis in [7],*

$$\begin{aligned} \dot{B}(x) &\leq 0 \qquad \forall x \in C, \\ B^+(x) - B(x) &\leq 0 \qquad \forall x \in D. \end{aligned}$$

Then, using $U = \mathcal{M}$, the conditions in (7) *hold. By Theorem 6.5, for each $r > 0$, $L_V(r)$ is forward pre-invariant for $\mathcal{H}$. Finally, note that $\mathcal{H}$ satisfies Definition 3.4. Then, as $C \setminus D$ is open, we have that for each $x \in C \setminus D$, $F(x) \in \mathrm{T}_C^{\mathcal{M}}(x) = \mathrm{T}_x\mathcal{M}$. As $L_V(r) \cap C$ is compact, Corollary 6.7 certifies that, for each $r > 0$, $L_V(r)$ is, indeed, forward invariant for $\mathcal{H}$. Figure 2 illustrates this result.*

VII. Conclusion

We present sufficient conditions for (strong) forward invariance of a closed set K for geometric hybrid dynamical systems. The proposed invariance certificates are formulated as nonstrict decrease conditions along solutions of a multiple barrier function, whose component scalar functions define K through the intersection of their zero-sublevel sets. These conditions are required to hold only in a neighborhood outside K. The barrier functions are required to be locally Lipschitz, and tools from nonsmooth analysis on manifolds are used to establish the results. Several running examples illustrate the results. Future work will focus on developing constructive methods for safe control design for classes of geometric hybrid systems evolving on manifolds.

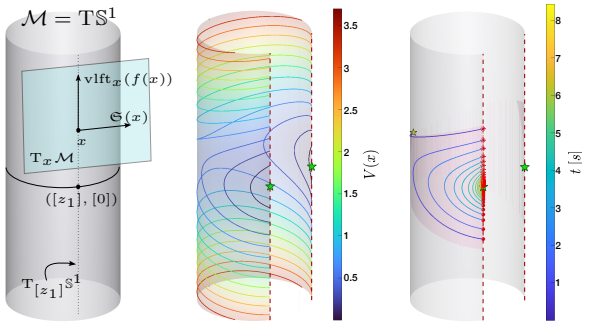


Fig. 2: *Left:* Construction of dynamics in Example 3.3, with $f(x) = -\nabla U([z_1]) - b[z_2]$. The cylinder denotes $\mathrm{T}\mathbb{S}^1$, where each 'fiber' is a tangent space to $\mathbb{S}^1$. *Center:* The contours represent level sets of the energy function V. The dashed red lines and the green stars denote points on the wall with $[z_2] \in \mathrm{T}_{[z_1]}\mathbb{S}^1$ and $[z_2] = 0$, respectively. *Right:* Simulated solution ϕ with initial condition denoted by a yellow star. The pink region represents the sublevel set $L_V(\phi(0,0))$. Observe that $\operatorname{rge}\phi \subset L_V(\phi(0,0))$.

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